

ASSIGNMENT 01

1. What is a List?
2. What is a Tuple?
3. What is the difference between List and Tuple?
4. Python Program to find the largest element in the list.
5. Python program to interchange first and last elements in a list.
6. Python program to swap two elements in a list
7. Python program to Reverse a List
8. Python program to count occurrences of an element in a list
9. Python program to find the sum of elements in a list
10. Python program to Multiply all numbers in the list
11. What are the ways to find the length of a list
12. Python program to find the smallest and largest number in a list (Without min-max function)
13. Python Program to find the area of a circle
14. Take inputs from the user to make a list. Again take one input from the user and delete that element from a list.
15. You are given a list of integer elements. Make a new list that will store a square of elements of the previous list. (With and without list comprehension)
 - `Input_list = [2,5,6,12]`
 - `Output_list = [4,25,36,144]`
16. WAP to create two lists, one containing all even numbers and the other containing all odd numbers between 0 to 151
17. Python program to Count Even and Odd numbers in a List
18. WAP to make new lists, containing only numbers which are divisible by 4, 6, 8, 10, 3, 5, 7, and 9 in separate lists for range(0,151)
19. Given a list containing elements of different types (integers, strings, and floats), create three separate lists to store integers, strings, and floats respectively.
20. What's The Difference Between The Python `append()` and `extend()` Methods?
21. Write a Python program to append a list to the second list
22. Write a Python program to find the third-largest number in a list
23. Write a Python program to check whether a list contains a sublist
24. Write a Python program to generate all sublists of a list
25. Write a Python program to find common items from two lists

26. How to flatten a list in Python?
27. How to sort a list in ascending and descending order without using the sort function?
28. How to sort a tuple?
29. Write a Python program to convert a list of multiple integers into a single integer
 - `[11, 33, 50] >>> 113350`
30. What is the difference between del and clear?
31. What is the difference between remove and pop?
32. What is the difference between indexing and Slicing?
33. What is the difference between sort and sorted?
34. What is the difference between reverse and reversed function?
35. What is the difference between copy and deep copy?
36. How to check whether the list is empty or not?
37. How to concatenate two lists?
38. Given a list of integers, create a new list where each element is replaced by the next largest number in the original list. If there is no larger number, replace the element with `-1`.
 - `input_list = [4, 2, 6, 8, 1, 2]`
 - `output_list = [6, 6, 8, -1, 2, -1]`
39. Write a function to reverse the elements within each tuple in a list of tuples.
 - `tuples = [(1, 2), (3, 4), (5, 6)]`
 - Output: `[(2, 1), (4, 3), (6, 5)]`
40. Write a function to add corresponding elements of tuples in two lists of tuples
 - `list1 = [(1, 2), (3, 4)]`
 - `list2 = [(5, 6), (7, 8)]`
 - Output: `[(6, 8), (10, 12)]`